# Chandra Vig - Learning Hub Guide

Companion to learning-hub/index.html. Systems to Optimus-adjacent robotics skills that redeploy. Visual learner; Poly C/C++ drivers (NXP, Qualcomm, Ambarella); avoid CUDA-deep as Plan A.

## How to open

cd /workspace/learning-hub && python3 -m http.server 8765
Open http://localhost:8765/ in Chrome/Firefox (or open index.html). No npm build required.

## Roadmap (6-12 months)

- M1-3: SoC/drivers + interview coding (Bootlin, char-driver mock, NeetCode C++)
- M4-6: ROS 2 / edge vision / PID (fake-ROS, perception pipeline, joint sim)
- M7-9: Host I/O / realtime (PCIe/DMA lab, war-story kit, Tier-1 apps)
- M10-12: Story + applications (SCCR war stories, 50/30/20 target mix)

**Weekly rhythm**

3x coding (45-60m, no AI on attempt); 2x systems/robotics visual; 1x build; 1x write-up

## Interactive labs (offline canvas)

**A SoC Factory Floor**

CPU foreman, DMA forklifts, IRQ bells, drivers, DRAM, ISP

**B Device Tree Binding**

Step: .dts -> match compatible -> probe -> /dev

**C Interrupt vs Polling**

Toggle mode; latency vs CPU busy; IRQ rate slider

**D PID Control Lab**

Kp/Ki/Kd sliders; cruise control / robot joint

**E ROS 2 Message Bus**

Nodes/topics; camera->cmd latency budget

**F Memory Hierarchy**

L1/L2/DRAM/disk hit-miss walk

**G PCIe/DMA Host Path**

BAR/MMIO control plane + DMA payload

## Guided tutorials

- Linux char driver mental model (open/read/write -> file_ops -> hardware)
- Bring-up checklist (power -> clocks -> DT -> probe -> userspace)
- Edge vision pipeline budget (sensor -> ISP -> encode/infer)
- Systems interview story template (SCCR)
- Interview coding warm-up rules (timer, no AI on attempt)

## Pet projects

**P1 Virtual Char Driver Lab**
projects/01-char-driver-lab/ -- C++ userspace mock + kernel stub; M1-3
**P2 Device Tree + Bring-up**
projects/02-dt-bringup/ -- sample .dts + validate_dt.py; M1-3
**P3 Mini Edge Perception**
projects/03-edge-perception/ -- OpenCV/stdlib pipeline + latency histogram; M4-6
**P4 ROS 2 Toy (fake ROS)**
projects/04-ros2-toy/ -- sensor/controller/visualizer pub-sub; M4-6
**P5 PID Joint Simulator**
projects/05-pid-joint/ -- Python + findings.md; M4-6
**P6 Host I/O War-Story Kit**
projects/06-host-io-stories/ -- DMA/IRQ/DT cases + lock_bench; M7-9
**P7 Interview Gym Tracker**
projects/07-interview-gym/ -- NeetCode + systems checklist; ongoing

## Resource priorities (hub library has verified URLs)

P0: Bootlin, LiveOverflow drivers, Device Tree talk, NeetCode, Educative C++ concurrency
P1: Branch SoC/CPU, Ben Eater, Nand2Tetris, CS:APP, Modern Robotics, Brian Douglas PID, ROS 2 tutorials
P2: Qualcomm Academy Robotics, Embedded CV Coursera, OpenCV Bootcamp, Ambarella DevZone (gated)
P3: Optional RDMA (Marvell lightboard / RoCE)

## Job targets

Tier 1: OpenAI Host Systems; NVIDIA networking/host; Broadcom NOS
Add: Embedded humanoid; Edge perception; Robot compute/host; ISP/sensor firmware; Real-time controls; Autonomy BSP

## North star

Tesla Optimus via redeployable skills only. Deepen SoC/drivers, add ROS/control/edge vision, keep coding sharp without AI crutches, treat CUDA as optional adjacency.